Asmita Datta

Marie Skłodowska-Curie Actions Joint Doctoral Fellow
University of Padova & University of Cyprus
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Education

PhD in Physics *Marie Skłodowska-Curie Actions Joint Doctoral Fellowship* - AQTIVATE — Oct 2023-Mar 2027 (expected)
Department of Physics and Astronomy (DFA) - University of Padova, Italy
- ***Supervisor:*** *Prof. Simone Montangero*

Department of Physics - University of Cyprus, Cyprus
- ***Supervisors:*** *Prof. Constantina Alexandrou and Prof. Haralambos Panagopoulus*

BS-MS Dual Degree *Physics (Major) | Data Science and Engineering (Minor)* — Aug 2018-July 2023
Indian Institute of Science Education and Research (IISER), Bhopal, India — GPA: 9.27/10
MS Thesis Title: Quantum machine learning for determining exchange-correlation potentials in many-body systems
- ***Supervisor:*** *Dr. Kuntal Roy*

Research Interests

My research focuses on quantum simulation, particularly tensor-network methods for studying the simulability of quantum algorithms. I am broadly interested in quantum computing and information, and the development of efficient methods for characterizing and certifying complex quantum systems.

Workshops and other Professional Training

Lattice Practices *The Cyprus Institute, Nicosia, Cyprus* — Sept 2024
Foundations of Quantum Computing *Royal Holloway University of London, UK* — Sept 2024
Summer School on Scientific Writing and Entrepreneurship *Vilnius University, Lithunia* — Jun 2024
Quantum Computing and Tensor Networks *University of Padova, Italy* — May 2024
Machine learning for science and engineering *TU Berlin, Germany* — Feb 2024
Exascale computing and scalable algorithms *KTH, Stockholm, Sweden* — Nov-Dec 2023

Secondments

CINECA *Bologna, Italy* — May-July 2025
DESY *Zeuthen, Germany* — March 2026

Publications

A. Datta, F. Baccari, I. Siloi and S. Montangero. *Benchmarking tensor network simulations of Shor's algorithm with coherent error propagation (In preparation)*, 2026.

F. Baccari, D. Bacilieri, M. Ballarin, F. P. Barone, F. Campaioli, A. G. Catalano, G. Cataldi, A. Coppi, A. Costantini, **A. Datta**, *et al. Quantum TEA: qtealeaves*, Version 1.5.8, Zenodo, 2025. doi:10.5281/zenodo.14883066.

Software Contributions

Quantum TEA : qmatchatea https://doi.org/10.5281/zenodo.11619265 (*next public version onwards*)

Research Experience

Quantum Information and Matter Group: **Doctoral Researcher** — Oct 2023-Present
- Benchmarked the performance of Quantum Fourier Transform (QFT) algorithm, characterising the propagation of errors and its scalability on a tensor-network based QC emulator (QMatcha TEA).
- Benchmarking Shor's algorithm on QMatcha TEA using MPS simulations of its quantum circuit, developing and comparing different performance metrics and parameters governing the overall quality and scalability of the

simulations, in the presence of intrinsic tensor-network truncation errors and extrinsic errors applied through a noise model with unitary over-rotations.

Quantinuum quantum circuit tensor network simulation challenge: Team member May-July 2025
Benchmarked the quantum chemistry suite of circuits as part of a broader team, using tensor networks. Our software ranked second overall. (Github Repository)

IISER, Bhopal: **MS Thesis (Chanakya Fellowship)** Aug 2022-May 2023
Quantum Machine Learning for determining exchange-correlation potentials in many-body systems (Thesis Link)
Developed ANN and Quantum Neural Networks (QNN) models to map the HSE xc potential range separation parameter ω based on material properties for 40 semiconducting and insulating materials. Generated training databases using VASP calculations and extracted material features via pymatgen from the Materials Project.

University of Freiburg, Germany: DAAD WISE Scholar May 2022-July 2022
Modelling Quantum Networks using three level atomic systems (Github Repository)
Modelled a quantum network with three nodes, each representing a three level atom, and studied the damage and reparation of the network by tuning parameters like transition rate, state preparation using pulses, and continuous wave driving frequency using Python and compared its resilience to an equivalent classical network.

London Mathematical Laboratory Summer School: Summer Intern July 2021
Tackling Quantum many-body systems using Artificial Neural Networks (Video Presentation)
Studied the 1D Quantum Ising and Heisenberg models using a RBM ansatz and simulated their ground state energies, spin-spin correlation functions, and magnetization using NetKet.

Technical Skills

Programming: C, C++, Python (advanced) | MATLAB, Mathematica, Java (basic)
Version control and other softwares: Gitlab, Github, VASP and LaTeX
High Performace Computing: Slurm

Academic Achievements & Recognitions

UNIPD Arqus Fellowship for Vilnius summer school. 2024
MSCA Doctoral Fellowship for PhD Research from AQTIVATE Project. 2023
Chanakya Fellowship by the I-HUB QTF, IISER Pune for MS Thesis. 2022
DAAD WISE Scholarship at University of Freiburg, Germany for Summer Research Insternship. 2022
MITACS Globalink Research Internship at University of Regina, Canada. (*Deferred for DAAD WISE*) 2022
NTU-India Connect Internship at NTU, Singapore. (*Deferred for DAAD WISE*) 2022

Conference Presentations - Talk & Poster

Contributed Talks

- Benchmarking tensor network simulations of Shor's algorithm with coherent error propagation
  AQTIVATE Conference, Larnaca, Cyprus Sept-Oct 2026
- Benchmarking Shor's algorithm on a tensor network-based quantum emulator
  A Panorama of Tensor Networks, Krakow, Poland Oct 2025

Poster

- Simulation of Shor's prime factorization algorithm on Quantum Matcha TEA
  Entangle This VI: 100 years of quantum, Madrid, Spain Sept 2025

Outreach & Science Communication

AQTIVATE Fellow Blog The art of compression: From cartography to quantum simulations. [Link] April 2026
World Quantum Day Event execution and disseminating quantum science among locals. 2025, 2026

More about me

Nationality: Indian | **Languages:** Bengali (Native), English (Fluent C2), Hindi (Fluent), Italian (Beginner)
Interests: Painting, Travelling and Baking